



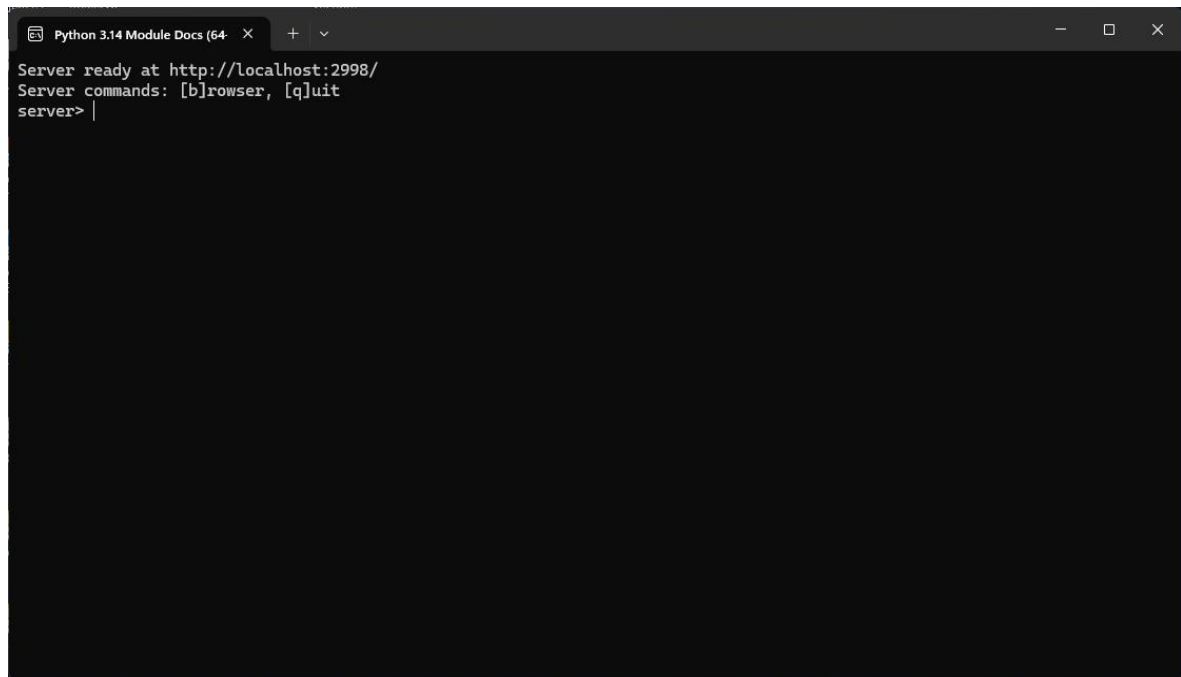
ADDIS ABABA UNIVERSITY
COLLEGE OF NATURAL AND COMPUTATIONAL
SCIENCE
DEPARTMENT OF MOLECULAR SCIENCE AND
GENETICS
MASTERS OF SCIENCE IN BIOINFORMATICS
ADVANCED PROGRAMMING FOR BIOINFORMATICS

ASSIGNMENT 1

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Submission date: April 24 2026
Submitted to: Nigatu (PhD)

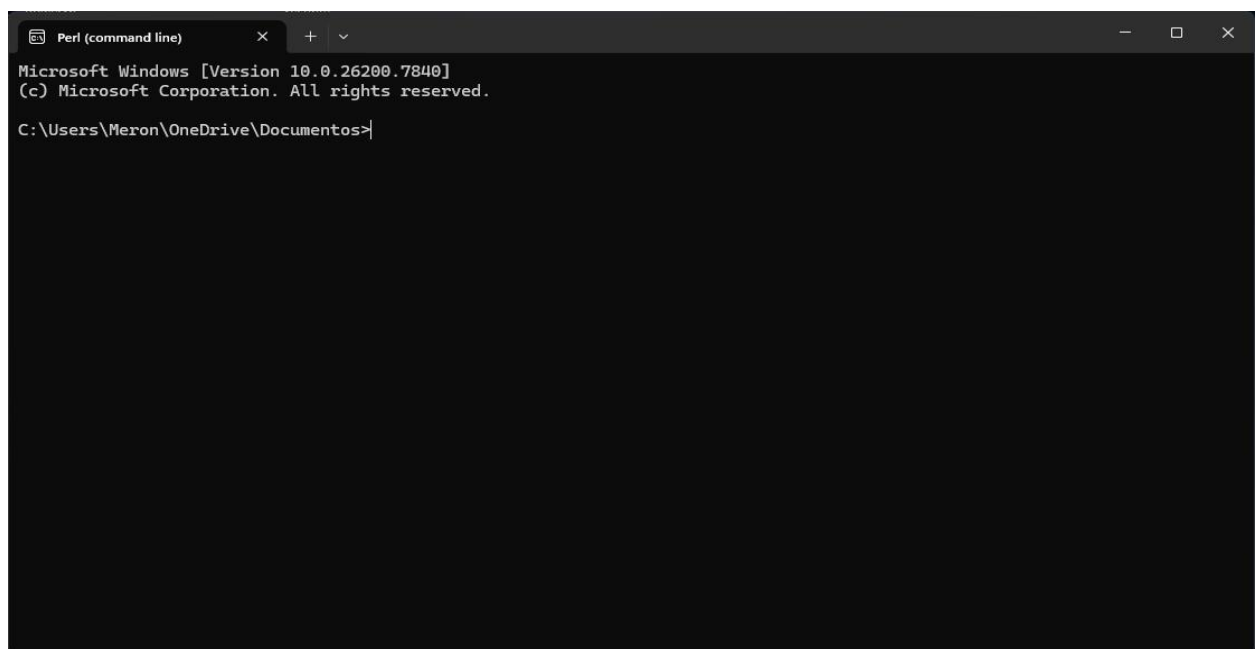
Task 1: Environment Setup

A screenshot of a web browser window titled "Python 3.14 Module Docs (64)". The browser shows a terminal interface for a Python web server. The text in the terminal reads: "Server ready at http://localhost:2998/", "Server commands: [b]rowser, [q]uit", and "server> |".

```
Python 3.14 Module Docs (64) × + -
Server ready at http://localhost:2998/
Server commands: [b]rowser, [q]uit
server> |
```

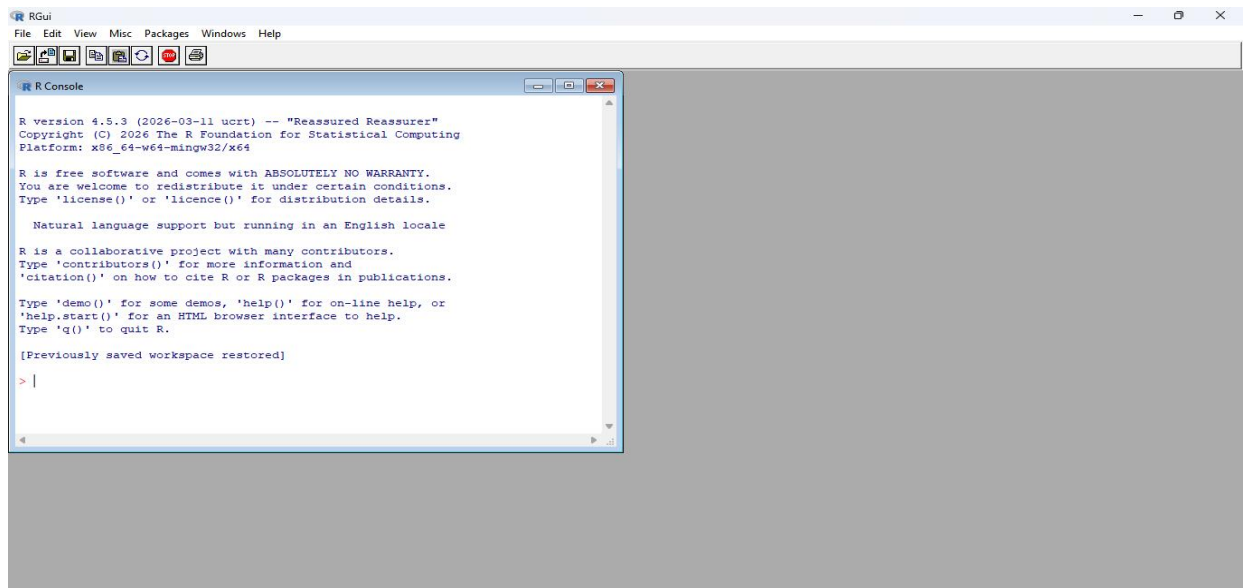
Version: Python 3.14

Python 3.14 is the latest stable release and introduces major improvements like template string literals, deferred evaluation of annotations, multiple subinterpreters, and a more colorful REPL. It focuses on developer experience, concurrency, and performance enhancements.

A screenshot of a Windows command prompt window titled "Perl (command line)". The window shows the standard Windows boot text: "Microsoft Windows [Version 10.0.26200.7840]", "(c) Microsoft Corporation. All rights reserved.", and the current directory path "C:\Users\Meron\OneDrive\Documents>".

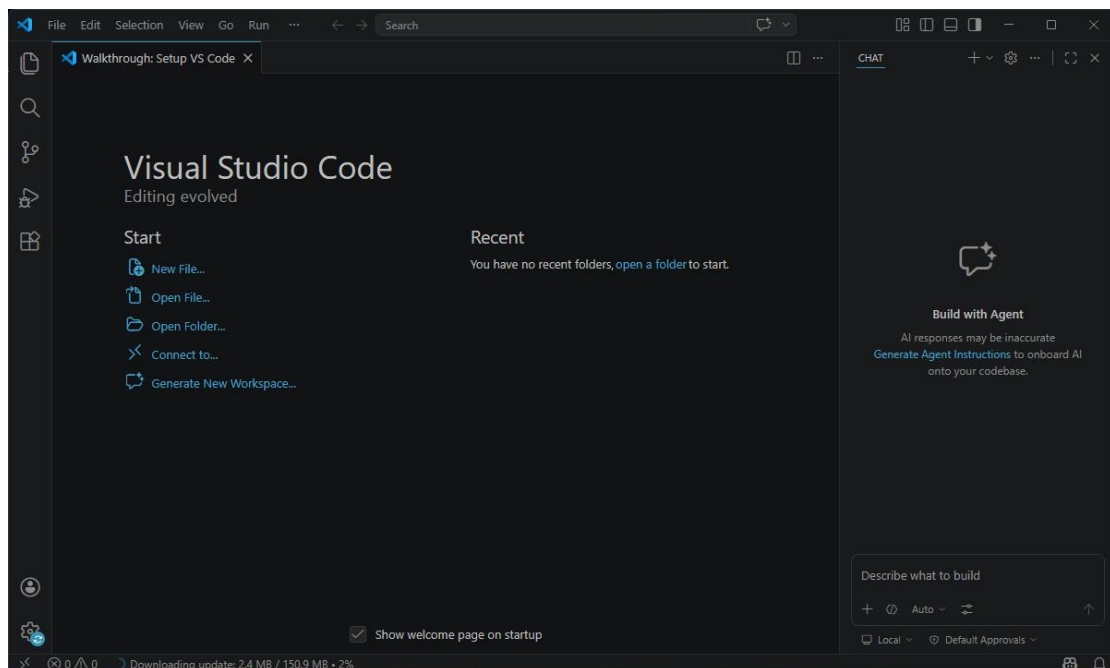
```
Perl (command line) × + -
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Meron\OneDrive\Documents>
```

Perl is a programming language designed for text processing and general-purpose scripting.



Version: R 4.5.3

R is a programming language and environment mainly used for statistics, data analysis, and visualization.



VS Code is a lightweight yet powerful editor for writing and debugging code. It supports many programming languages and has a rich ecosystem of extensions to customize your workflow.

Task 2: Linux File System Navigation

pwd

```
2026-04-22 16:26.01 /home/mobaxterm pwd
/home/mobaxterm
```

ls

```
2026-04-22 16:26.13 /home/mobaxterm ls
Desktop LauncherFolder MyDocuments README.txt
```

mkdir bioinfo_lab

```
2026-04-22 16:28.25 /home/mobaxterm mkdir bioinfo_lab
```

cd bioinfo_lab

```
2026-04-22 16:30.38 /home/mobaxterm cd bioinfo_lab
```

touch sample.txt

```
2026-04-22 16:34.16 /home/mobaxterm/bioinfo_lab touch sample.txt
```

cp sample.txt copy.txt

```
2026-04-22 16:55.25 /home/mobaxterm/bioinfo_lab cp sample.txt copy.txt
```

mv copy.txt renamed.txt

```
2026-04-23 11:29.27 /home/mobaxterm/bioinfo_lab mv copy.txt renamed.txt
```

Questions:

1. What does **pwd** do?

The command **pwd** stands for “**print working directory.**”

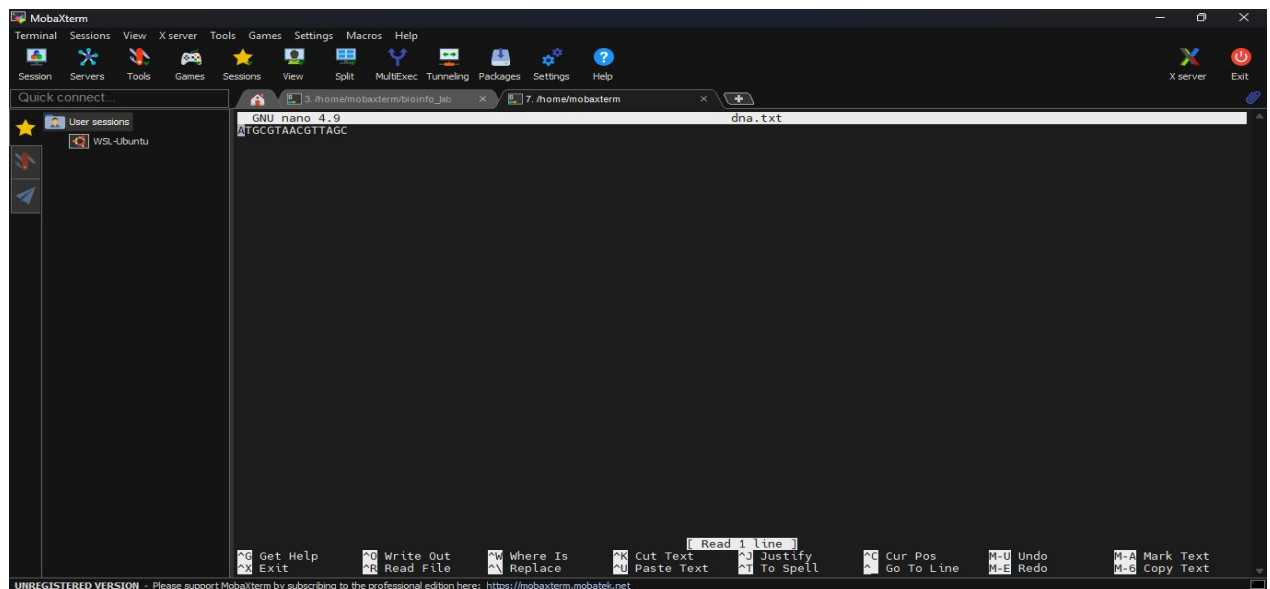
When `pwd` is typed in a Unix/Linux or macOS terminal, it shows the full path of the directory you are currently in.

2. Difference between `cp` and `mv`?

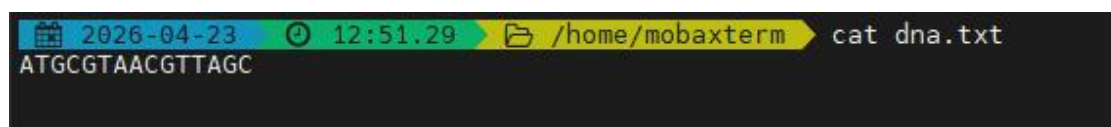
`cp` - copy- Makes a duplicate of a file or folder. The original stays in place, and a new copy is created at the target location.

`mv` - move - Transfers a file or folder to a new location. The original is removed from its old place and only exists in the new location.

Task 3: File Handling & Text Processing



Editing a file called `dna.txt` in the terminal using the nano text editor. Inside the file, I typed a DNA sequence (ATGCGTAACGTTAGC).



This shows the command `cat dna.txt` in the terminal. The `cat` command simply prints out the contents of a file, so the output you see (ATGCGTAACGTTAGC) is the DNA sequence stored inside `dna.txt`

```
2026-04-23 12:51.40 /home/mobaxterm grep "CGT" dna.txt
ATGCGTAACGTTAGC
```

This shows the command **grep "CGT" dna.txt** in the terminal. The grep command searches for a specific pattern of text inside a file, in this case, the DNA sequence "CGT". The output shows the line from dna.txt that contains that pattern (ATGCGTAACGTTAGC).

Questions:

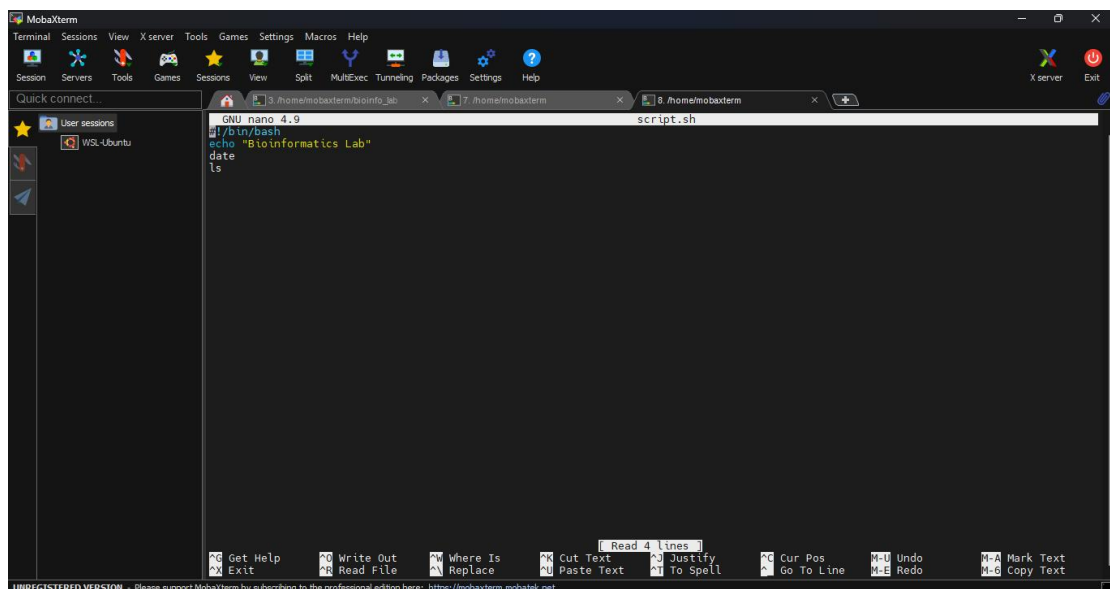
1. What does **grep** do?

The grep command in Linux/Unix is used to search for specific text patterns inside files and display the matching lines. It's one of the most powerful tools for quickly finding words, phrases, or regular expressions in large amounts of text.

2. What is the output of the search?

All the outputs are shown above

Task 4: Shell Scripting



```
GNU nano 4.9 script.sh
#!/bin/bash
echo "Bioinformatics Lab"
date
ls
```

A is a Bash script being edited in the nano text editor.

- ✓ **#!/bin/bash** → This tells the system to use the Bash shell to run the script.
- ✓ **echo "Bioinformatics Lab"** → Prints the text *Bioinformatics Lab* to the terminal.

- ✓ date → Displays the current date and time.
- ✓ ls → Lists all files and directories in the current folder.

```
2026-04-23 19:56.54 /home/mobaxterm chmod +x script.sh
```

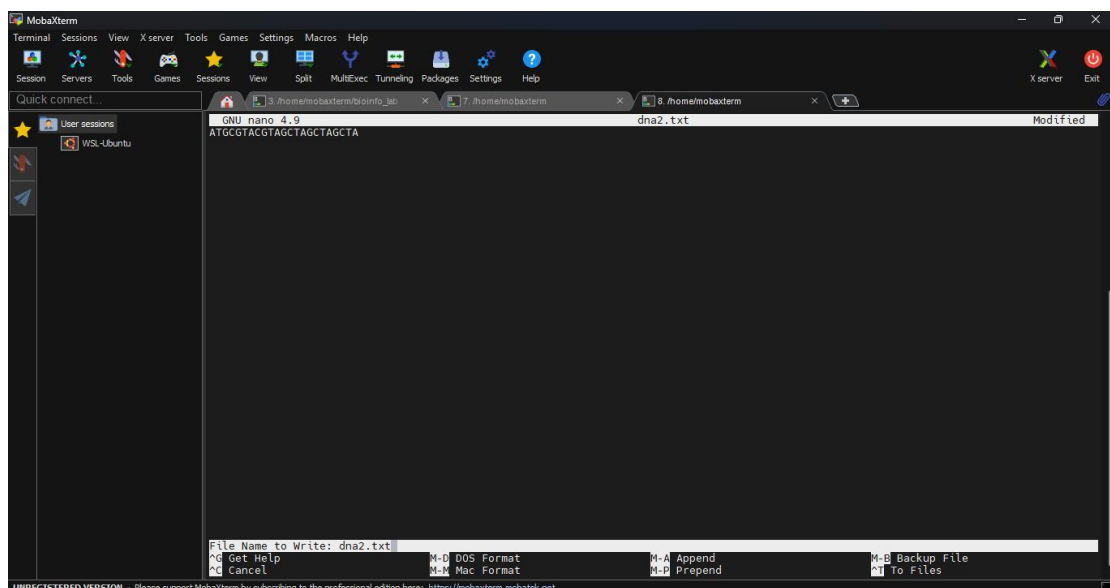
- ✓ chmod → short for *change mode*, it's used to modify file permissions in Linux/Unix.
- ✓ +x → adds the “executable” permission, which allows the file to be run like a program.
- ✓ script.sh → the file you're applying this permission to (the Bash script).

By running this command, it made the script.sh file executable. That means instead of calling it with `bash script.sh`, it can now run it directly.

```
2026-04-23 19:57.03 /home/mobaxterm ./script.sh
Bioinformatics Lab
Thu Apr 23 20:03:21 EAST 2026
Desktop LauncherFolder MyDocuments README.txt bioinfo_lab dna.txt nano.save script.sh
```

This shows what the script runs after making it executable. The output first prints “Bioinformatics Lab” (from the echo command), then shows the current date and time (from the date command), and finally lists all files and folders in the working directory (from the ls command).

Task 5: Simple Bioinformatics Task



This shows editing and saving a file called **dna2.txt** in the nano text editor. Inside the file, there is typed a DNA sequence (ATGCGTAGCTAGCTAGCTAGCTA).

```
2026-04-23 20:11.54 /home/mobaxterm wc dna2.txt
1 1 24 dna2.txt
```

wc stands for *word count*. It shows three things

- ✓ Lines (1) → how many lines the file has.
- ✓ Words(1) → how many words are inside.
- ✓ Characters(24) → the total number of characters (including letters, numbers, spaces, etc.).

```
2026-04-23 20:25.23 /home/mobaxterm grep "A" dna2.txt
ATGCGTACGTAGCTAGCTAGCTA
```

The command searched for the letter “**A**” in the file(dna2.txt) and printed the line where it was found.